Final Architecture

PDF edition of preserved repository document | docs/final_architecture.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final architecture

The reusable core stores entities, relations, observations, goals, states, actions, events, results and trajectories as typed immutable records. The generic structural Environment protocol defines reset/state/goal/legal-actions/step/goal-reached/trajectory, parameterized by observation/state/action/transition/trajectory types; both current environments satisfy its runtime structural check. Dataset backends implement generation, canonical identity, strict verification and evaluation annotation; the shared runner does not implement arithmetic. Current State includes decimal-specific fields, so arbitrary domains would require an adapter or a generalized payload schema. This is an honest boundary of reuse.

Multiplication has two retained environments: historical exact digit mechanics and controlled numerical claims. In the latter, the environment exposes a hand-designed long-multiplication cursor and accepts all 100 numerical claims. It writes the learner's proposed units/carry even when numerically wrong. Only the private reference generator, training-feedback boundary and post-prediction verifier compute the expected claim. Inference never gets a corrected digit or numerical verifier feedback.

Model-visible state uses operand digits, cursor, lengths, current accumulated digit, carry, phase and static goal/relation flags. There is no product or future correct state. The final adapter passes indexed numerical features through a frozen SmolLM2 transformer and concatenates a categorical encoding of those same features. A two-layer learned head predicts decimal answer digits for conventional A/structured conditions or units/carry for procedural B/C. All share one checkpoint revision, backbone and head capacity; training masks only inactive policy heads. This is frozen-feature task adaptation, not full language-model fine-tuning. The language model's indexed inputs limit how much natural-language pretraining capability is being tested.

Working memory follows actual transitions. Episodic SQLite memory stores training-only own attempted experiences and binary feedback; C reuses only accepted own actions. Concept memory stores descriptive text and does not answer arithmetic. Its presence alone is not a measured benefit. The matched-information conventional local policy uses B's exact predictor/state labels and real execution but bypasses the memory wrapper.

The bounded dream engine creates tagged hypothetical states using proposed-state dynamics, never real environment evidence. Its arithmetic scoring is model confidence, not a correctness oracle. The generic integer-position goal demo separately demonstrates goal-directed bounded rollout; it is not a learned second-domain experiment. Final decisions tag real/hypothetical results, record100 candidates and proposals, and receive correct-action/transition judgments only after prediction.

Final artifacts include strict dataset manifests, head NPZ checkpoints, backbone revision metadata, configs, full predictions, structured evidence, failure groups, resource counters and statistics. The frozen model remains local and offline after its initial download. The old NumPy MLP and all failed studies are preserved.

The running final study loaded its original annotation class at protocol freeze. Real decisions have inline posthoc correct-action judgments. No-interaction decisions additionally receive immutable hypothetical_judgments.json sidecars after evaluation; raw predictions are retained. The current backend also supports those posthoc hypothetical annotations for fresh reproductions. Neither route supplies oracle values to inference or promotes hypothetical agreement to real trajectory validity.